

Proxmox VE Greenfield Cluster Build & Troubleshooting Guide

Purpose. A complete procedure for standing up a multi-node Proxmox VE cluster from bare hardware, including the decisions that shape it and the failures most likely to be encountered. Written to be followed by someone who has not done it before.

Reader. A practitioner executing a build, not a reader evaluating one. Steps are imperative. Rationale for every non-obvious choice is stated in the same section as the step it governs.

Reference build. Three nodes, mixed hardware, single flat VLAN, no shared storage, one administrator. Values from that build appear as worked examples and are marked as such. Substitute your own.

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Companion case study evanireland.tech/projects/proxmox-cluster/

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Sequence

The order below is not arbitrary. Two dependencies drive it, and both are explained where they occur: a node holding any guest cannot join a cluster, and a node joining a cluster has its configuration replaced.

Step	Work	Blocking constraint
1	Pre-flight verification	Network and firmware settings are far cheaper before installation than after
2	First node installation	-
3	Post-install hygiene	Package repositories must be corrected before any update succeeds
4	Remaining node installation	-
5	Cluster formation	Must occur before any guest exists on any node
6	Storage configuration	Must occur after cluster formation
7	Access & identity	-
8	Backup infrastructure	Requires storage
9	Verification	-

Pre-Flight Verification

Network Reachability

The installer accepts a static IP address without testing it. A node configured on a network that has no router comes up isolated, and the failure is not visible until first boot.

Confirm before installing. Attach a laptop to the switch port the nodes will use, assign a static address in the intended subnet, and verify three things in order:

```
ping <gateway>          # a router exists in this subnet
ping 1.1.1.1             # that router forwards off-subnet
nslookup proxmox.com     # name resolution works
```

All three must pass. A failure at the first indicates a VLAN configured on the switch but not on the router - a layer-2 broadcast domain with no gateway in it. Nodes on such a network reach each other and nothing else.

VLAN Delivery

Two configurations are possible and they impose different requirements.

Untagged access port. Everything downstream sees plain Ethernet. No tagging is configured on the host and `vmbr0` remains a plain bridge. Required where downstream equipment cannot be relied upon to pass 802.1Q tags - powerline adapters and unmanaged switches in particular.

Trunk port. The host requires a VLAN-aware bridge and the management interface pinned to a VLAN. Guests can then be tagged onto any VLAN the trunk carries.

I would recommend untagged access ports for installation regardless of the eventual design. Converting to a trunk afterwards is a configuration change; recovering an unreachable node is a trip to the machine.

The consequence of untagged delivery must be understood before the build, not discovered after it: guests on those nodes cannot be tagged onto any other VLAN. Any service that must reach a different network reaches it by routing at the firewall, or by an internal bridge with NAT. Both are viable. Neither is free.

Firmware Settings

Hardware virtualization is the only non-negotiable entry. Without it the hypervisor does not function.

AMD platforms:

Setting	Value	Function
SVM Mode	Enabled	AMD-V hardware virtualization. Ships disabled on ASUS boards.
IOMMU	Enabled	Address translation for PCIe passthrough
Above 4G Decoding	Enabled	Maps device memory above the 4 GB boundary; prerequisite for passthrough
ErP Ready	Disabled	Preserves standby power to the network interface, retaining Wake-on-LAN
CSM	Disabled	Forces UEFI boot
Secure Boot	Disabled	Supported by Proxmox but adds no benefit here
Restore AC Power Loss	Power On	The node rejoins the cluster unattended after an outage

Intel platforms: substitute Virtualization Technology (VTx) for SVM Mode and Virtualization Technology for Directed I/O (VTd) for IOMMU. The remaining entries are identical in function.

Memory Configuration

Memory capacity dominates memory speed for virtualization workloads. The difference between JEDEC default timings and a rated XMP/DOCP profile is single digit percent for running guests; the difference between a stable host and an unstable one is total.

I would recommend leaving memory at JEDEC defaults on any host where the rated profile has not been proven stable. Four dual-rank modules populating all slots is the hardest load a consumer memory controller can be given, and rated speeds across four modules frequently fail to POST. Where a rated profile is applied, run at minimum one full MemTest86 pass before trusting it.

Reference build: 64 GB DDR4-3600 across four modules was left at 2133 MT/s deliberately.

Drive Health Baselines

Record a health reading for every drive before it holds anything. One reading is a number; two readings are a trend, and the second reading is worthless without the first.

Spinning disks and SATA SSDs:

```
smartctl -a /dev/sdX      # attributes and firmware revision
smartctl -t long /dev/sdX # extended self-test; hours on a spinner, minutes on an SSD
smartctl -l selftest /dev/sdX # result
```

SATA SSD attributes carry different meaning than hard disk attributes:

Attribute	Meaning
177 Wear_Leveling_Count	Normalized value counts down from 100. Approximate percent of rated write endurance remaining
241 Total_LBAs_Written	Lifetime writes. $\text{LBAs} \times 512 \text{ bytes} / 10^{12} = \text{terabytes written}$, against the drive's rated TBW
179 Used_Rsvd_Blks_Cnt_Tot	Spare blocks consumed. A climbing value indicates degradation in progress
199 UDMA_CRC_Error_Count	Indicates a cable or port fault, not a drive fault. Healthy drives are replaced over this attribute
5 Reallocated_Sector_Ct	Zero expected

NVMe reports health directly and requires no self-test:

```
smartctl -a /dev/nvme0
```

Critical Warning expected 0x00. Percentage Used is a direct wear estimate. Available Spare must remain well above its threshold. Media and Data Integrity Errors expected zero. Unsafe Shutdowns disproportionate to Power Cycles indicates a machine that was habitually powered off uncleanly - not a fault, but useful history.

Firmware revision is worth recording. Specific drive models have shipped with firmware defects corrected in later revisions.

Installation Media

Obtain the current Proxmox VE ISO and verify its checksum before writing it:

```
Get-FileHash -Algorithm SHA256 .\proxmox-ve_<version>.iso
```

A truncated or corrupted image produces installation failures that present as hardware faults and consume hours.

Write the image in DD mode. In Rufus, select the ISO, and when prompted choose **Write in DD Image mode**. The partition scheme selector does not apply in DD mode - the image is written byte for byte including whatever partition table it already carries.

Ventoy will not work. Proxmox is a documented Ventoy failure case; the installer does not boot from its hybrid layout.

Verification that DD mode was used: insert the completed stick into a Windows machine and open Explorer. If Windows cannot read it and offers to format it, DD mode succeeded. If a normal FAT32 volume full of files is browsable, the image was written in ISO mode and must be rewritten.

First Node Installation

Node Selection

Install the node that will **create** the cluster first, and install it on the hardware most tolerant of being wiped and redone.

The creating node's `/etc/pve` is the copy that survives cluster formation. Every joining node has its own replaced. The node that will accumulate configuration first should therefore be the node that creates the cluster.

Installer Options

Target disk. Select deliberately. On a machine with more than one disk the installer offers all of them, and selecting the wrong one destroys data with no confirmation beyond the summary screen. Where a machine holds a data disk that must survive, physically disconnecting it for the duration of the installation is the cheapest available insurance.

Filesystem. Under Options, `ext4` produces an LVM volume group containing a thin pool.

I would select ext4 with LVM-thin over ZFS on any single-disk node. ZFS's defining capability is self-healing from redundancy. On a single disk there is nothing to heal from, and what remains is checksums that detect corruption which cannot then be repaired. ZFS's ARC cache additionally defaults to consuming up to half of system memory, which is unacceptable on a low-memory host. LVM-thin already provides snapshots and thin provisioning.

This choice is reversible without reinstalling. Proxmox replication requires ZFS *storage*, not a ZFS *root*. A ZFS pool can be created on a secondary disk later.

Volume sizing. Accept the defaults. Note that `maxroot` **does not follow the `hdsiz/4` formula stated in older documentation** - the installer caps it substantially lower.

Reference build: a 930 GB drive produced a 96 GB root, an 8 GB swap volume, and a 794 GB thin pool.

The `local` storage lives on the root filesystem. Its capacity is therefore the root volume's capacity. This constrains what may be stored there and is addressed under Storage Configuration.

Network configuration. The management interface must be the **wired** adapter. Proxmox does not configure wireless networking. Where no interface is listed at all, the network driver did not load and the installation should not proceed.

The hostname must be a fully qualified domain name containing at least one dot, and must not resolve to `127.0.0.1`.

Renaming a node after it has joined a cluster is a documented but disruptive manual procedure. Treat the hostname as permanent.

Use a syntactically valid administrative email address. Proxmox passes it to Postfix for alert delivery, and an invalid address produces continuous mail system errors in the logs.

Verification

The web interface is served on **port 8006** over HTTPS with a self-signed certificate. The browser warning on first access is expected.

Authenticate as `root` with realm **Linux PAM standard authentication**. Selecting the wrong realm is the most common first-login failure.

The subscription notice is informational, not an error.

Expected state: web interface reachable, root login succeeds, and `ping 1.1.1.1` returns from `Node > Shell`.

Post-Install Hygiene

Package Repositories

A fresh installation is configured against the **enterprise** repository, which requires a paid subscription key and returns HTTP 401 without one. The first `apt update` fails.

The **no-subscription** repository carries the same packages with less staging. A third repository, `pvetest`, carries pre-release packages and should not be used.

Via `Node > Updates > Repositories`: disable the component `pve-enterprise`, disable the enterprise Ceph repository, then add **No-Subscription**.

The enterprise Ceph repository produces the same 401 whether or not Ceph is in use.

Proxmox 9 runs on Debian 13 and uses the **deb822** format - `.sources` files with `Types:` / `URIs:` / `Suites:` / `Components:` / `Signed-By:` fields, replacing the single-line entries of earlier releases.

Confirm:

```
cat /etc/apt/sources.list.d/*.sources
```

System Update

```
apt update
apt full-upgrade
```

`apt update` refreshes the package index and installs nothing.

Use `full-upgrade`, never `upgrade`. `full-upgrade` is permitted to remove packages where a dependency change requires it. `apt upgrade` refuses to remove anything and silently holds back kernel and Proxmox package transitions. This is the most common route to a partially upgraded hypervisor whose faults appear months later.

Reboot where a new kernel was installed. `uname -r` reports the running kernel; where it differs from the newest installed kernel, the new kernel is present but not in use.

Virtualization Verification

Firmware settings are confirmed at the operating system, not assumed from the BIOS screen.

```
lscpu | grep -i virtual          # AMD-V or VT-x
lsmod | grep kvm                # kvm_amd or kvm_intel
dmesg | grep -i -e "AMD-Vi" -e DMAR -e IOMMU # IOMMU initialization
```

`AMD-Vi` and `DMAR` are the AMD and Intel identifiers for the same subsystem. Absence of the virtualization line indicates the firmware setting did not take effect and no guest will start.

Time Synchronization

Cluster membership is time-sensitive. Nodes with divergent clocks produce faults that present as network problems. This is verified before cluster formation, not after.

```
timedatectl
```

Expected: correct time zone, `System clock synchronized: yes`, `NTP service: active`.

Where an upstream router intercepts NTP and answers locally, this is acceptable and arguably preferable for a cluster - **mutual agreement between nodes matters more than absolute accuracy against UTC**, and a single local source guarantees agreement.

The negative case is worth recording: Active Directory rejects Kerberos tickets beyond approximately five minutes of clock skew. Where a domain controller is later hosted on the cluster, time synchronization is the first thing to examine on a Kerberos fault, not the last.

Storage Sanity

```
lsblk
pvesm status
```

Two storages are expected: `local`, a directory storage on the root filesystem, and `local-lvm`, the thin pool for guest disks.

Remaining Node Installation

Repeat installation and hygiene on each remaining node. Only the target disk, hostname, and IP address differ.

Configure nothing that resides in `/etc/pve` on any node until the cluster exists. This includes storage definitions, users, firewall rules, and backup jobs. The reason is stated under Cluster Formation.

Confirm `timedatectl` reports synchronized on every node before proceeding.

Cluster Formation

Timing

A node holding any guest cannot join a cluster. `pvecm add` refuses. The joining node must be emptied first.

Form the cluster while every node is empty. Deferring cluster formation does not defer its cost - it creates one, because every guest built in the interim must be destroyed or manually migrated before its node can join.

This contradicts the common advice to cluster only once the layout is settled. That advice correctly observes that leaving a cluster is considerably harder than joining one. It is nonetheless outweighed by the mechanical constraint above.

Configuration Replacement

`pvecm add` replaces the joining node's `/etc/pve` in its entirety with the cluster's copy. Storage definitions, users, firewall rules, and backup jobs on the joining node are destroyed.

The creating node's configuration is the copy that survives. This is the only respect in which the creating node differs from the others - all Proxmox cluster members are peers afterwards.

Pre-Flight

On every node:

```
qm list      # QEMU/KVM guests - must be empty
pct list     # LXC containers - must be empty
hostname -f  # must return the fully qualified name
```

`hostname -f` returning a bare hostname or an error indicates an incorrect `/etc/hosts`, which produces cluster faults.

Creation

On the node selected to create the cluster:

```
pvecm create <clustername>
```

The cluster name is written into `corosync.conf`. **Keep it short, use letters, digits and hyphens only, avoid dots, and do not reuse a node name.** Changing it later is a manual and disruptive procedure.

The node becomes a single-member cluster, quorate and fully functional.

```
pvecm status
```

Joining

Retrieve the join information from the creating node: `Datacenter > Cluster > Join Information > Copy Information`. The encoded string carries the cluster name, the peer address, and the certificate fingerprint.

On each joining node: `Datacenter > Cluster > Join Cluster`, paste, supply the **creating node's** root password.

The command-line equivalent is run **from** the joining node **at** an existing member:

```
pvecm add <ip-of-existing-member>
```

Join one node at a time. Allow each join to complete and verify before starting the next. A partially failed join is difficult to unwind.

Three behaviors during a join are normal and should not prompt troubleshooting:

1. A warning that existing configuration will be lost. This is the `/etc/pve` replacement.
2. The joining node's web session breaking. Joining replaces the node's self-signed certificate with one issued by the new cluster certificate authority. A fresh browser warning, a brief 401, or a connection error follows. Wait, hard-refresh, authenticate again.
3. Automatic SSH key exchange between members. No manual key work is required.

Quorum Verification

```
pvecm status
```

Field	Expected	Meaning
Nodes	node count	Members currently visible
Expected votes	node count	Votes configured
Total votes	node count	Votes currently present
Quorum	$\text{floor}(n/2)+1$	The threshold
Flags	Quorate	The cluster is functional. Absence of this word is the entire problem whenever one exists

```
corosync-cfgtool -s
```

Reports corosync link status. One link is expected where one network exists.

Quorum Behavior

A three-node cluster has a quorum of two. One node may be powered off freely. **A second node becoming unavailable while the first is off drops the cluster below quorum: running guests continue to run, but `/etc/pve` becomes read-only and no guest may be started, stopped, or reconfigured until a second node returns.**

This is the operational cost of a three-node design and it should be understood before it occurs.

Migration Capability

Live migration requires compatible CPU architectures between source and destination. Nodes of the same processor generation support live migration. **A node of a different vendor or a materially different generation supports offline migration only** - a live migration will fail on CPU feature mismatch.

Corosync Network Separation

Proxmox recommends a physically separate network for corosync, so that cluster heartbeats do not compete with guest and storage traffic.

Where each node has a single network interface this is unavailable. **The compromise is acceptable where all nodes share a quiet local switch and the traffic classes that could starve corosync - backups and migrations - are themselves node-to-node on that same switch.** Where spurious quorum loss appears, the remedy is a second corosync link on a second interface, not timeout tuning.

High Availability

Leave `Datacenter > HA` empty unless shared storage and redundant networking are both present.

Without HA, loss of quorum renders `/etc/pve` read-only. Running guests continue. The condition is self-correcting when a node returns.

With HA enabled, a node that loses quorum triggers a **watchdog self-fence** - it reboots itself to guarantee it is not running a guest another node has assumed ownership of. On a cluster with a single corosync link, or one where a node is deliberately powered off, HA converts a routine and harmless condition into unexpected reboots.

Storage Configuration

Storage is configured after cluster formation, for the reason stated above.

Root Filesystem Protection

`local` is directory storage residing on the root filesystem. By default it accepts VZDump backup files.

Remove VZDump from `local` 's content types on every node. A backup job directed at `local` fills the root filesystem, and a full root filesystem on a hypervisor produces failures that present as anything except a full disk.

`Datacenter > Storage > local > Edit`, content set to ISO image, Container template, and Snippets only.

Storage Type Selection

The choice is mechanical rather than a matter of preference.

Requirement	Storage type
ISO images, container templates, VZDump backups	Directory only. No other type can hold them
Guest disks, block level	LVM-thin, ZFS, Ceph
Guest disks as files	Directory

A directory storage is therefore required wherever ISOs, templates, or backup files must live.

Directory Storage Creation

Proxmox refuses to use a disk carrying existing partitions. Wipe first via `Node > Disks > [disk] > Wipe Disk`. **Confirm the model and capacity in the dialog before accepting.** This is the irreversible step of the procedure.

A single 16 MB partition on an otherwise empty disk is a **Microsoft Reserved Partition**, created when Windows initializes a disk as GPT. Its presence indicates the disk was initialized under Windows and holds no data partition.

`Node > Disks > Directory > Create: Directory` writes a partition table, creates a filesystem, mounts it under `/mnt/pve/<name>` by way of a systemd mount unit, and registers it in the cluster storage configuration - in one action.

Node Restriction

`storage.cfg` is cluster-wide. A storage backed by a local disk on one node, left unrestricted, is advertised to every node. Nodes without that disk report the storage as failing, permanently.

`Datacenter > Storage > [storage] > Edit` :

- **Nodes:** the owning node only
- **Shared:** unchecked. A local disk is not shared storage
- **Content:** as required

A storage restricted to a node reports **disabled in pvesm status** on nodes it does not belong to. This is correct behavior and not a fault - it is precisely the outcome the restriction exists to produce.

Tiering

Where a node holds both solid-state and mechanical storage, **store templates and ISO images on the mechanical disk and run guests from the solid-state pool.** Templates are large and read rarely. Running a guest directly from a mechanical disk is performance-limited for no benefit.

Access & Identity

Administrative Account

Proxmox maintains a user database independent of Linux accounts, with realms, roles, and access control lists.

Create a user in the **Proxmox VE authentication server** realm and grant it a role at path `/` with propagation.

Role selection. `PVEAdmin` is frequently recommended and is **wrong for a single-administrator cluster**. It explicitly excludes `Sys.Modify`, `Sys.PowerMgmt`, `Realm.Allocate`, and `Permissions.Modify`. Creating a datacenter backup job requires `Sys.Modify`. Rebooting a node from the interface requires `Sys.PowerMgmt`. The result is an account that cannot perform routine administration.

I would assign Administrator where one person administers the cluster. Least privilege means matching privilege to the job performed. Where the job is *administer this cluster*, that is the correct role. Restricted roles earn their place where a junior technician should manage guests but never networking.

The limit of this control should be stated plainly: `Administrator` includes `Sys.Console`, so the account can open a root shell on a node through the web interface. It is not a containment boundary. What it provides is a separately revocable credential, an audit trail distinguishing routine work from break-glass work, and an account that can carry two-factor authentication.

Where containment is genuinely required, a custom role comprising `Administrator` minus `Sys.Console` provides it.

Proxmox users reside in `/etc/pve` and are therefore cluster-wide on creation.

Two-Factor Authentication

Enable TOTP on the administrative account and generate recovery keys. Store the recovery keys outside the cluster.

Leave the `root@pam` account without two-factor authentication as a break-glass credential. Two-factor on every administrative account is stronger until a lost authenticator device locks the administrator out of the hypervisor entirely. This trade is acceptable specifically where the web interface is not internet-facing; it should be reconsidered where it is.

SSH

Generate an Ed25519 key pair and protect it with a passphrase. Load it into an SSH agent so the passphrase is entered once per session rather than once per connection.

On Windows the agent service is present but disabled by default:

```
Get-Service ssh-agent | Set-Service -StartupType Automatic
Start-Service ssh-agent
ssh-add $env:USERPROFILE\.ssh\id_ed25519
```

In a Proxmox cluster, `/root/.ssh/authorized_keys` is commonly a symbolic link into `/etc/pve/priv/`. Where this is the case, a key added on one node is present on all of them. Verify rather than assume:

```
ls -la /root/.ssh/authorized_keys
```

The Proxmox installer sets `PermitRootLogin yes`, overriding the Debian default of `prohibit-password`. All nodes therefore accept root password authentication over SSH until changed.

Whether this warrants changing depends entirely on exposure. Root password authentication over SSH is dangerous where reachable from the internet, where automated attacks are continuous. On a management network with no inbound path, an attacker must already be inside the network, at which point a larger compromise has occurred. **Where it is left in place, the control that matters is a long and unique root password on every node.**

Close it where any inbound path is introduced - a port forward, a remote access VPN, or a mesh VPN subnet router advertising the management subnet:

```
sed -i 's/^PermitRootLogin yes/PermitRootLogin prohibit-password/' /etc/ssh/sshd_config
sshd -t # validates configuration; silence indicates success
systemctl restart ssh # or ssh.socket where socket-activated
```

`sshd_config` is per-node and is **not** part of the replicated cluster filesystem. Repeat on every node.

Retain the existing SSH session and verify with a second one before closing the first. A live session survives an `sshd` restart. Where a change is faulty, that session is the only remaining route in. The same discipline applies to firewall changes.

The fallback where SSH is lost entirely: **the web interface node Shell does not use SSH for the node being administered.** It remains available.

Host Shell Administration

Proxmox nodes are appliances. `pvecm`, `qm`, `pct`, `pvesm`, and any operation touching `/etc/pve` require root privilege. **An unprivileged account with `sudo` adds ceremony without limiting anything, because the account requires full root to perform its function.**

The audit-trail argument for `sudo` is real where multiple technicians administer a shared host, and near-worthless where one person does. **The correct answer differs by context and should be decided on that basis rather than by habit.**

Services within guests are a separate question and the standard practice applies without qualification: services run as unprivileged service accounts. Packaged software generally handles this already.

Container Security Defaults

Create containers unprivileged. In a privileged container, container root maps to host UID 0 - an escape yields the hypervisor. Unprivileged containers map container root to an unprivileged host UID.

The unprivileged flag cannot be changed in place. Conversion requires backup and restore with the flag set, or recreation.

Enable the Nesting feature for containers running systemd 257 or later - Debian 13 and equivalent. The creation task logs a warning to this effect. Modern systemd expects to create nested namespaces and mounts, which an unprivileged container cannot do without it.

`searchdomain` takes a **domain name**, not a resolver address. Supplying an IP address causes unqualified lookups to append it, producing nonsense queries.

I would not run a host firewall inside single-purpose service containers. Where a perimeter firewall filters the network and the service restricts access at the application layer, a third packet filter inside the container adds no coverage and adds a layer that will be forgotten during a future fault. Where host-level filtering is wanted, the Proxmox cluster firewall is the better location - centrally visible, resident in `/etc/pve`, and captured by configuration backups.

Backup Infrastructure

Placement

Backups must not reside on the node holding the data they protect, nor on a node that is routinely powered off.

Proxmox Backup Server may be installed directly onto a Proxmox VE node. **I would install it as a virtual machine instead.** Backups sharing an operating system with the node they protect means a single failed upgrade or a filled root filesystem removes both simultaneously. A virtual machine boundary costs a small amount of memory and produces a backup server that survives its host being rebuilt.

Disk Layout

Two virtual disks. A modest operating system disk and a separate datastore disk.

A datastore that fills does not then wedge the operating system, and the datastore can be grown, moved, or reattached independently. The recovery path becomes: reinstall the backup server on a new operating system disk, reattach the datastore disk, re-register the datastore. **The backup data survives the operating system being rebuilt entirely** - which is also why the backup server should be excluded from its own backup job. A backup of the backup server, stored on the datastore, is lost with the datastore.

Thin Pool Capacity

Where both disks are allocated from a thin pool, sum the allocations against the pool size and retain headroom. Thin provisioning consumes only what is written, but **an overcommitted thin pool that fills corrupts every guest resident on it simultaneously**. The headroom is not slack.

Least Privilege Credentials

Create a dedicated backup server user and grant it `DatastoreBackup` on the datastore, not `DatastoreAdmin`.

DatastoreBackup permits creating and reading backups. **It does not permit pruning or deletion.** The credentials held on each hypervisor can therefore add to backup history and cannot destroy it. Where a node is compromised, the attacker cannot reach through the backup connection and remove restore points - the mechanism by which ransomware most commonly defeats backups.

The consequence must be configured for: retention cannot be enforced by the client. Leave retention unset in the Proxmox VE backup job, where it would attempt a prune it lacks permission for and report failure. Configure the prune schedule on the backup server instead.

Backup Target Registration

Datacenter > Storage > Add > Proxmox Backup Server requires the server address, the datastore name, credentials, and the server's **fingerprint**, obtained from the backup server dashboard.

Leave this storage unrestricted by node. Unlike a local disk, a network-reachable backup server is available to every node. Node restriction addresses reachability, not tidiness.

Job Scope

Scope backup jobs by node where node roles are meaningful. A job scoped to the node holding persistent services, with selection mode set to **All**, requires no maintenance - every service subsequently built on that node is protected from the day it is created.

The alternatives each carry a recurring obligation. Pool-based selection requires adding every new guest to the pool. All-nodes-with-exclusions requires excluding guests that are created and destroyed routinely. **Both are the same failure pattern: a default that must be remembered.**

The gap in node-scoped selection should be documented rather than left implicit: a persistent service placed on an unscoped node is not protected.

Prune, Garbage Collection & Verification

Three schedules are required and each does a distinct job.

Prune marks snapshots for removal according to a retention policy.

Garbage Collection frees the disk space. **Prune alone does not reclaim capacity.** A datastore with a prune schedule and no garbage collection schedule fills while the interface reports that old backups have been removed.

Verification re-reads every chunk and validates it against its stored checksum. **Without it, the backups have never been confirmed readable.** "A backup exists" and "a backup works" are separate claims and only one of them has value during a restore.

Notification

Configure a notification target and direct job failures to it.

The default notification target delivers to the local root mailbox, which is not read. **A backup job that fails silently is worse than no backup job, because decisions are then made in the belief that protection exists.**

Restore Testing

A backup that has never been restored is an assumption. Perform one restore to a non-production target and record the result.

Troubleshooting Reference

DNS Interception

Symptom. A recursive resolver returns SERVFAIL for every query, including queries that should succeed. Query times indicate genuine network activity before failure.

A test returning the expected result for the wrong reason is worse than a failing test. Where every query fails, a DNSSEC test expecting SERVFAIL appears to pass and proves nothing.

Diagnosis, first step. Separate resolution failure from validation failure:

```
dig @<resolver> -p <port> example.com +cd
```

`+cd` sets the Checking Disabled bit, performing recursion while skipping DNSSEC validation.

- NOERROR - recursion works; the fault is in validation
- SERVFAIL - recursion itself is failing

Diagnosis, second step. Query a root server directly:

```
dig @198.41.0.4 . NS +norecurse
```

`198.41.0.4` is `a.root-servers.net`. `+norecurse` requests an answer from the server's own authority.

A correct response and an intercepted response differ in three independent ways:

Indicator	Intercepted	Correct
Status	REFUSED	NOERROR
Flags	qr ra	qr aa - authoritative, no ra
Answer records	0	13 root nameservers
Round-trip time	Implausibly fast for the distance	20-80 ms
Message size	Tens of bytes	800+ bytes

The `ra` flag is conclusive on its own. `ra` signals recursion available. **Root servers are authoritative-only and never set it.** A response bearing `ra` to a `+norecurse` query did not come from a root server.

Cause. A router or security appliance transparently redirecting outbound port 53. The resolver's queries to authoritative servers are answered by a forwarding resolver that holds no root zone, so recursion fails at its first step.

Resolution. Exempt the resolver's address from the appliance's DNS interception features. Many such products offer per-device exemption, which resolves the fault without altering DNS behavior for any other host.

Sequencing note. DNS interception features are commonly documented as affecting clients only. **This is incomplete: a recursive resolver's own outbound queries are DNS traffic.** Interception starves the resolver itself, so the exemption is a prerequisite for the resolver functioning at all, not a step in a later client cutover.

A DNSSEC validation pass, once resolution works:

Query	Expected	Proves
A correctly signed name	NOERROR with the ad flag set	ad is Authenticated Data. It is set only after a DNSSEC chain validates successfully
A deliberately broken name	SERVFAIL, or no answer at all	Invalid data is refused

A timeout rather than SERVFAIL on the broken name remains a pass. The resolver continues seeking a chain that validates before formally giving up, and the client stops waiting first. Confirm by repeating the query with `+cd`: an immediate cached answer with validation disabled, and no answer with it enabled, isolates the refusal to validation.

Authentication Failure Versus Reachability Failure

Read the failure mode before forming a hypothesis.

A firewall block never produces an authentication prompt. A dropped connection times out; a rejected connection returns `Connection refused`. **Reaching a password prompt proves the connection succeeded, the protocol handshake completed, and the service offered authentication.** The fault is at the application layer.

"Cannot reach it" and "reached it and was refused" are different problems. Distinguishing them from the error text halves the search space.

Common cause of root SSH authentication failure: Debian ships `PermitRootLogin prohibit-password`. Root may authenticate by key only, but the server still prompts for a password and then denies it.

```
grep -REi '^s*(PermitRootLogin|PasswordAuthentication)' \
/etc/ssh/sshd_config /etc/ssh/sshd_config.d/
```

Recursion is required because modern OpenSSH also reads drop-in files, where a setting overrides the main file.

Correct resolution is a key, not enabling password authentication.

SSH Key Rejected Without Explanation

The server ignores a key file whose permissions permit access by other users, falls back to password authentication, and reports only "Permission denied."

```
chmod 700 ~/.ssh
chmod 600 ~/.ssh/authorized_keys
```

Permission Check Failed

`Permission check failed (/, Sys.Modify) (403)` indicates the authenticated user's role lacks the named privilege at the named path.

`Datacenter > Permissions > Roles` lists every role with its complete privilege set. Consult it rather than inferring from the role name.

An account lacking `Permissions.Modify` cannot repair its own permissions. Authenticate as `root@pam` to correct role assignments.

Configuration Validation Before Restart

Several services provide a configuration parser that reports errors without affecting the running service. Use them before every restart.

Service	Command	Success indication
unbound	<code>unbound-checkconf</code>	no errors in <file>
OpenSSH	<code>sshd -t</code>	Silence

A syntax error caught by a validator costs seconds. The same error caught by a failed restart is a service outage during which the fault must be located.

Verification Checklist

Infrastructure is complete when the next person to build a service on it does not first have to repair the platform.

Cluster

- ☐ `pvecm status` reports the expected node count and `Flags: Quorate`
- ☐ `corosync-cfgtool -s` reports all links connected
- ☐ Every node visible in a single web interface
- ☐ `Datacenter > HA` empty, unless shared storage and redundant networking are present

Nodes

- ☐ Enterprise repositories disabled, no-subscription enabled, `apt update` clean
- ☐ `full-upgrade` completed and rebooted where the kernel changed
- ☐ Hardware virtualization confirmed at the operating system on every node
- ☐ `timedatectl` synchronized on every node

Storage

- ☐ VZDump removed from `local` content types on every node
- ☐ Local storages restricted to their owning node, Shared unchecked
- ☐ `pvesm status` clean on every node

Access

- ☐ Administrative account created with an adequate role and two-factor authentication
- ☐ Recovery keys generated and stored off the cluster
- ☐ Break-glass account documented with a strong unique password
- ☐ SSH key authentication verified to every node

Backup

- ☐ Backup server reachable, datastore created
- ☐ Least-privilege credentials in use, retention configured server-side
- ☐ Backup job created, **executed manually at least once**, and the resulting snapshot confirmed present
- ☐ Prune, garbage collection, and **verification** schedules all configured
- ☐ Failure notifications delivered to an address that is read
- ☐ At least one restore performed and recorded

Records

- ☐ Health baseline recorded for every drive
- ☐ Address plan documented
- ☐ Configuration copy stored off the cluster
- ☐ Deferred decisions recorded with the condition that would reverse them

Deferred Decision Register

A decision deferred without a recorded trigger is a decision forgotten. Each entry states the condition that reverses it rather than a review date.

Deferred	Condition that reverses it
Root password authentication over SSH	Any inbound path: port forward, remote access VPN, or a mesh VPN subnet router advertising the management network
Single corosync link	Recurring spurious quorum loss
Undersized backup datastore	Protected data approaching the datastore's deduplicated capacity
ext4 over ZFS	A requirement for cross-node replication, which needs ZFS storage on both ends
No off-site backup copy	Any data on the cluster whose loss would be material

Attribution

Written by Claude (Anthropic) in collaboration with Evan Ireland, from a live three-node build executed 27-30 August 2026. Procedures were executed and verified on that build rather than compiled from documentation. Failures documented in the Troubleshooting Reference were encountered during it.